Cosmic Signature

White paper 1.3 | A procedural art protocol on Arbitrum One

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White paper 1.3 | Canonical English edition | Proposed contract V3

The main description below specifies the proposed V3 implementation at commit `9d1a6447` on branch `v3.1-2026-10-13`. V2 remains deployed in the chain snapshot used here. Publication of this paper does not activate V3. The deployment history and version comparisons are in Sections 10 and 11; Appendix A identifies the deployed contracts.

Abstract

Cosmic Signature connects timed participation on Arbitrum One with procedural art. Participants make **Gestures** using ETH or CST, the protocol's ERC-20 token. Gestures extend a **Performance Cycle**, establish time-based recognition, and record participation in **Stellar Selections**. A successful finalization distributes configured portions of the ETH reserve, creates Cosmic Signature NFTs with seeds recorded on-chain, and schedules the next cycle. The remaining reserve carries forward; growth from one cycle to the next is not guaranteed.

V3 changes how participation is recognized. Participation CST accrues linearly with elapsed time and is created for the previous participant when the next Gesture arrives. Participant selections use the ETH base cost at each Gesture as a weight. A late-cycle premium increases the cost of acting close to the deadline without increasing that Gesture's selection weight. V3 also changes the ETH allocation split, expands the main NFT allocation, and replaces the CST window's duration adjustment with an adjustable decline rate.

The contracts determine timing, ownership, allocation, and seed provenance. Artwork is computed off-chain by an open-source three-body simulation and renderer. Reproduction depends on the generator version and its inputs as well as the seed. The protocol remains upgradeable and owner-configurable within the constraints described below; neither governance transfer nor removal of owner powers happens automatically in V3.

1 Design and terminology

The design combines three things: a public clock that participation can extend, explicit rules for distributing a shared ETH reserve, and artwork derived from recorded seeds. ETH Gestures add to the reserve. CST Gestures consume CST instead of adding ETH. Finalization links the completed cycle to a new set of NFTs.

The two currencies have different cost schedules. ETH base costs step upward with ETH participation after the opening Gesture. CST base costs decline with time, while participation adjusts the decline rate. These mechanisms create different ways to participate; the code does not enforce a fixed ETH-to-CST exchange rate or guarantee a balanced mix of the two currencies.

The paper uses reader-facing names. The Solidity interfaces retain older names, so the following mapping helps readers inspect the implementation.

In this paper	In the contracts
Gesture	`bidWithEth`, `bidWithCst`, and related methods
Performance Cycle	`roundNum` and the current round state
Cycle Finalization Time	`mainPrizeTime`
Signature Allocation	The main allocation distributed by `claimMainPrize`
Stellar Selection	The participant and anchored-NFT selection functions
Anchoring	The two `StakingWallet` contracts
Imprint and Retrieve	Token creation and asset-retrieval operations
Cosmic Council	`CosmicSignatureDao`
Public Goods Vault	`CharityWallet`
Outreach Reserve	`MarketingWallet`
Allocations Wallet	`PrizesWallet`

All ETH and CST arithmetic uses integer smallest units: one ETH and one CST each contain `10^18` units. Time in transaction logic is measured in seconds, except the stored cycle increment, which is in microseconds. Division of positive integers rounds down unless stated otherwise. Numeric defaults describe the reviewed configuration, not immutable promises about every future cycle.

2 System architecture

An existing UUPS proxy is the protocol's stable interaction address. V3 replaces its implementation while retaining its storage. The implementation inherits common and V2 base contracts for much of its behavior; the V3-specific source files alone are not the entire protocol.

Component	Responsibility
Protocol proxy and implementation	Gesture pricing, the cycle clock, ETH reserve, participation records, and finalization
CST token	ERC-20 balances, protocol-authorized creation and burns, holder-authorized burns, and delegated voting checkpoints
Cosmic Signature NFT	ERC-721 ownership, stored seeds and names, creation events linking NFTs to cycles, and configurable metadata references
Random Walk NFT	Existing external collection used for the ETH reduction and a separate Anchoring track
Allocations Wallet	Secondary ETH balances, attached NFTs, cycle beneficiary records, and retrieval permissions
Per-cycle token holders	Custody of attached ERC-20s, controlled through the Allocations Wallet
Anchoring wallets	NFT custody and eligibility for ETH distributions or anchored-NFT selections
Public Goods Vault and Outreach Reserve	Configured ETH forwarding and treasurer-directed CST distribution
Cosmic Council	Proposals and execution using delegated CST voting power

The website and indexer help people discover state and prepare transactions. They do not determine what a transaction does. Similarly, storing a seed on-chain does not store the rendered image or video on-chain. Generator execution, metadata publication, and media hosting are separate services.

An implementation upgrade does not redeploy the token, NFT collections, Council, or other companion contracts. Their existing code and state remain in effect unless a separate deployment or configuration transaction changes the relevant component. This distinction matters particularly for the Allocations Wallet; Section 10.4 describes its optional replacement.

3 The Performance Cycle

3.1 Opening a cycle

A cycle becomes available at `roundActivationTime`. Its first Gesture must use ETH, optionally with the Random Walk NFT reduction. That first Gesture starts the countdown and the CST Calibration Window. A CST Gesture cannot open a cycle.

Except for the original cycle's fixed opening base cost of 0.0001 ETH, an ETH Calibration Window determines the opening base cost. It starts at a stored ceiling and decreases to a floor. At default parameters the ceiling is twice the preceding cycle's opening **undiscounted base cost**, and the floor is `floor(ceiling / 200) + 1 wei`. The next ceiling does not include a swallowed overpayment or the Random Walk NFT reduction.

The descent uses the following integer interpolation, where `C` is the ceiling, `F` the floor, `u` elapsed seconds since activation, and `T` the window duration:

```
opening base cost = C                              when u <= 0
opening base cost = C - floor((C - F) * u / T)     when 0 < u < T
opening base cost = F                              when u >= T
```

Here `T = floor(I / ethDutchAuctionDurationDivisor)`, where `I` is the stored increment in microseconds. At the initial one-hour increment the duration is approximately two days. If no one opens the cycle, the opening cost remains at the floor after the window elapses, unless the owner changes the applicable configuration. After the window has elapsed and before the first Gesture, the owner can use `halveEthDutchAuctionEndingBidPrice` to lower the floor further; that operation changes both the floor divisor and window duration divisor.

3.2 Extending the clock

The opening Gesture sets the deadline to the current timestamp plus `floor(I / initialDurationUntilMainPrizeDivisor)`, approximately one day at initial settings. Every subsequent Gesture adds `floor(I / 1,000,000)` seconds to the **stored deadline**. It does not reset the deadline to an interval after the latest transaction.

The increment began at one hour. On finalization it becomes `I + floor(I / mainPrizeTimeIncrementIncreaseDivisor)`. The default divisor is 100, giving approximately 1% growth per completed cycle, subject to rounding. There is no fixed maximum cycle length.

An expired countdown does not close participation. Until finalization succeeds, another Gesture can become the Final Gesture and extend the stored deadline. If the deadline was far enough in the past, it can remain expired after that extension. A participant can then make a Gesture and finalize in the same transaction through a contract. This behavior is present in V2 and retained in V3.

3.3 Finalization

The Final Gesture participant can finalize once the deadline has arrived. Other addresses can finalize after an additional timeout, currently 48 hours in the V2 snapshot and retained by V3 initialization. The successful caller becomes the cycle beneficiary and receives the Signature Allocation. When that caller differs from the Final Gesture participant, the endurance and Chrono-Warrior calculations still use the participation history.

Finalization updates the recognition records, distributes allocations, creates the NFTs and fixed CST allocations, and prepares the next cycle. It is an explicit transaction; merely reaching a timestamp does not execute it. Failed finalization reverts the transaction's state changes and transfers.

The next activation is scheduled from the finalization timestamp using `delayDurationBeforeRoundActivation`. Its Solidity default is 30 minutes, but the checked V2 deployment uses **seven days**. V3 initialization preserves that live setting. The owner can change this delay separately.

4 Gesture costs and participation

4.1 ETH Gestures

Every ETH Gesture, including the opening Gesture, sets the next ETH base cost to `base + floor(base / ethBidPriceIncreaseDivisor) + 1 wei`. The default divisor is 100. CST Gestures do not advance this ETH base-cost sequence.

The amount due is the base cost plus any late-cycle premium. A Random Walk NFT owner can use an eligible NFT for a reduction to `ceil(amount due / 2)`. The NFT stays with its owner; the protocol marks its ID as used for this reduction permanently. Transferring it does not restore eligibility. This one-time reduction and the NFT's one-time Anchoring eligibility are separate records.

ETH sent above the amount due is refunded when the excess exceeds `ethBidRefundAmountInGasToSwallowMaxLimit * tx.gasprice`. Otherwise the excess stays in the reserve and is recorded as paid. The default threshold parameter is 6,843 gas units. A refund failure reverts the Gesture. Neither the excess nor the late premium increases the next base cost or selection weight.

4.2 The CST Calibration Window

Let `B` be the current window's beginning cost, `a` its beginning timestamp, `m` the decline rate in CST smallest units per second, and `t` the current time. The V3 base cost is:

```
CST base cost = max(0, B - (t - a) * m)
```

The late-cycle premium is added to that base. The full paid amount is burned from the participant's CST balance. The participant specifies a maximum allowed CST cost; the transaction reverts if the amount due exceeds it. Protocol-directed burning does not require an ERC-20 spending approval to the protocol.

The decline rate changes on Gestures. With change divisor `d` (default 100):

```
ETH candidate rate = m + floor(m / d)
CST updated rate   = floor((m + 1) * d / (d + 1))
```

Every ETH Gesture, including the opening Gesture, attempts the rate increase and adopts its candidate only if `candidate * 180 <= B`. After the opening Gesture has started the CST window, further ETH Gestures do not restart it: the updated rate applies to the entire elapsed interval `t - a`, so an accepted

increase can immediately lower the CST base cost. A CST Gesture adopts its updated rate (which integer rounding can leave unchanged) and restarts the window at:

```
new B = max(2 * premium-free CST base cost,
            configured beginning-cost minimum,
            updated rate * 180)
```

The configured minimum is set to 1 CST by V3 initialization. The additional rate-based floor gives a restarted window a theoretical duration of at least three minutes. It is **not a cooldown**: another Gesture can arrive immediately, and the safeguard does not promise three minutes of remaining time after an ETH rate adjustment. The first window of a later cycle can be shorter because its carried-over beginning cost was established at an earlier rate.

The beginning cost created by the first CST Gesture in a cycle is carried into the next cycle's first CST window. The decline rate also carries between cycles. If a cycle has no CST Gesture, its carried beginning cost stays unchanged. V3 initialization preserves V2's existing carried beginning cost of 200 CST; it does not immediately replace that first-window cost with 1 CST.

At sufficient elapsed time the base cost reaches zero. A zero-cost CST Gesture remains valid, and a zero base also produces zero late premium. Gas and other transaction requirements still apply. Participation is not subject to a one-Gesture-per-second restriction.

4.3 The late-cycle premium

V3 makes late participation more expensive without allowing the premium to increase selection weight or ratchet later base costs. It applies to ETH and CST Gestures after the opening Gesture.

The window length is `L = floor(I / D)`, where `D` is `roundLateBidDurationDivisor`, initialized to 3,000,000. At the original one-hour increment this is 20 minutes; with the checked live increment it is 1,236 seconds. The window grows as the increment grows.

For positive base cost `p`, elapsed time `e` within that window is clamped between zero and `L`. The exact implementation uses these integer operations:

```
e       = clamp(L - (deadline - now), 0, L)
q       = floor(e * M / I)
premium = floor(q^k * p / 2^(13*k))
due     = p + premium
```

The implementation skips the premium calculation when `e` is zero or no Gesture has yet opened the cycle; in either case the premium is zero.

Here `M = 29,228,998,656` and `k = 8` at V3 initialization. The approximate shape is `due / p = 1 + 4 * (e / L)^8`. The exact integer formula is authoritative. The table shows the smooth approximation, rather than exact smallest-unit amounts from a particular cycle.

Fraction of late window elapsed	Approximate total cost relative to base
0%	1.00 times
50%	1.016 times
75%	1.40 times
90%	2.72 times
100% and overtime	5.00 times

For example, with a 1 ETH base cost, the exact total at the deadline is 5.000050302627095474 ETH at the original one-hour increment. At the snapshot increment of 3,709,083,600 microseconds, it is 4.993485445647992173 ETH. The integer window length and intermediate rounding account for the difference; five times the base is an approximation, not an exact fixed multiplier.

The Random Walk NFT reduction applies after adding the premium. The premium is a one-time cost: subsequent ETH base costs, restarted CST base costs, and selection weights use premium-free values. Total paid-amount statistics include the premium. Inside the window, the amount due may change between a quote and execution; in particular, the CST amount due need not decrease monotonically.

4.4 Messages and attached assets

A Gesture can include a message of up to 280 bytes under the default setting. Bytes are not necessarily characters. Combined methods also accept an attached ERC-20 amount or ERC-721 NFT. Approval must be granted to the contract that actually transfers the attached asset, the Allocations Wallet.

Attached assets stay separate from the ETH reserve. The Allocations Wallet holds attached NFTs; a dedicated token holder for each cycle holds attached ERC-20s. The cycle beneficiary has the first right to retrieve them after finalization. The `donateEth` and `donateEthWithInfo` methods can also add ETH to the reserve without creating a Gesture, selection weight, or Participation CST. A plain ETH transfer with empty calldata instead invokes the protocol's receive function and attempts an ETH Gesture with default arguments.

5 CST creation and supply

5.1 Participation CST goes to the previous participant

Suppose Alice makes a Gesture and Bob makes the next one. Bob's transaction creates Participation CST for **Alice**, based on the elapsed time between the two Gestures. This is true whether either Gesture uses ETH or CST. If Alice makes both Gestures, the same address receives the new CST. When CST is used, the cost is burned before the Participation CST is created; an anticipated creation cannot substitute for the balance needed to pay that cost.

For elapsed seconds `s`, multiplier `R`, and increment `I` in microseconds:

```
Participation CST, in smallest units = floor(s * R / I)
```

V3 initializes `R` to `60,000,000,000,000,000,000,000,000` (`6 * 10^25`). At the original one-hour increment, this corresponds to 1 CST per elapsed minute.

Time between Gestures	CST created for the previous participant at the original increment
Same timestamp	0
1 minute	1
1 hour	60
1 day	1,440

These are examples at the original increment, not a fixed rate for all cycles. The checked deployment's larger increment would give approximately 0.971 CST per minute after V3 initialization. Increasing the increment reduces the rate unless the multiplier is separately adjusted.

The opening Gesture creates no Participation CST. Finalization does not settle an additional Participation CST amount for the Final Gesture participant, so elapsed time alone does not produce a claimable balance. The existing minimum Participation CST argument now constrains the amount created for the previous participant, not an amount promised to the sender; V3 skips that check on the opening ETH Gesture.

5.2 Fixed allocations and burns

At default settings, finalization creates 1,000 CST for each recognition or NFT-selection allocation and 3,000 CST for the Outreach Reserve. The main allocation receives 1,000 CST **in total**, even though V3 gives it three NFTs. A typical cycle therefore still creates 27,000 CST in fixed allocations, plus Participation CST created during the cycle. Section 6 explains omitted tracks.

The token starts with zero supply. There is no protocol schedule that fixes its future total supply or guarantees an inflationary or deflationary outcome. Supply changes reflect Participation CST, fixed allocations, and burns. Holders can also burn their own CST, and approved spenders can use the inherited allowance-based burn function. Protocol-authorized creation and burns use the token's immutable protocol address.

There is a technical supply ceiling: the inherited `ERC20Votes` implementation rejects creation that would raise supply above `2^208 - 1` CST smallest units. This representation limit is not a scheduled economic supply cap or a promise about the amount that will circulate.

6 Reserve distribution and recognition

6.1 ETH allocation

The reserve is the protocol's ETH balance, including retained ETH and direct contributions. Finalization computes each allocation against that balance before making its outgoing transfers. V3 initialization sets these percentages:

Track	Share	Destination
Signature Allocation	20%	Successful finalization caller
Chrono-Warrior	15%	Recipient's balance in the Allocations Wallet
Participant ETH Stellar Selection	5% total	Three equal allocations in the Allocations Wallet
Cosmic Signature Anchor Distribution	5%	Cosmic Signature Anchoring Wallet
Public Goods	5%	Configured Public Goods destination
Retained reserve	Remainder	Protocol contract for later cycles

The configured outgoing percentages sum to 50%. Actual outgoing ETH is affected by integer rounding and exceptions: if no Cosmic Signature NFTs are anchored, the Anchor Distribution stays in the protocol; if the Public Goods transfer fails, that amount also stays. Splitting the participant ETH allocation can leave additional wei in the reserve. Retention is not a promise that later cycles start with a larger reserve.

The Signature Allocation is transferred directly to the caller. A caller that cannot receive ETH causes finalization to revert. Secondary ETH recipients do not have to receive ETH during finalization because their amounts are escrowed. The Public Goods transfer has a specific failure-tolerant path; that does not make every external call during finalization failure-tolerant.

6.2 NFT and CST allocation

Track	Cosmic Signature NFTs	CST
Signature Allocation	3	1,000 total
Endurance Champion	1	1,000
Chrono-Warrior	1	1,000
Final CST Gesture, if present	1	1,000
Participant NFT Stellar Selection	10	10 allocations of 1,000
Anchored Random Walk NFT selection, if eligible NFTs exist	10	10 allocations of 1,000
Outreach Reserve	0	3,000

With all tracks present, a V3 cycle creates **26 NFTs and 27,000 fixed CST**. If there was no CST Gesture, subtract one NFT and 1,000 CST. If no Random Walk NFTs are anchored at finalization, subtract ten NFTs and 10,000 CST. Multiple allocations can go to the same address. NFT and CST counts are counts of allocations, not necessarily distinct recipients.

The table uses the V3 main NFT count and the retained standard allocation settings shown in Appendix B. Those configurable counts and amounts can change.

NFTs and CST are created directly for recipients during finalization. The main allocation's NFTs have consecutive IDs; its event reports the first ID and count. Token creation does not invoke ERC-721 receiver callbacks. A contract address receiving an NFT must therefore have its own means to manage it.

6.3 Endurance Champion and Chrono-Warrior

The Endurance Champion recognizes the Gesture that remains the most recent for the longest interval. Chrono-Warrior recognizes the Gesture whose Endurance Champion record then stands for the longest interval. These are related but different measures: a long quiet interval can establish a new record late in the cycle, while an earlier, smaller record may have stood longer.

The implementation evaluates records when later Gestures or finalization run. It reconstructs when a new record overtook the previous one instead of simply starting the reign at the transaction that notices it. Strict comparisons and second-level rounding determine ties; repeated Gestures by one address remain distinct events. The final quiet interval is considered at finalization, which can change the recipients then. A delayed finalization can therefore affect recognition. The endurance and Chrono-Warrior specification provides the exact algorithm and boundary examples.

6.4 Weighted Stellar Selections

Each Gesture receives a weight equal to the **undiscounted, premium-free ETH base cost quoted at that moment**, in wei. This rule applies to ETH Gestures, ETH Gestures with the Random Walk NFT reduction, and CST Gestures, including zero-cost CST Gestures. Weight is not the actual amount paid in either currency.

At finalization, the implementation samples the cumulative weights to choose the three participant ETH allocations and ten participant NFT allocations. Selections allow repeats. Under uniformly distributed random inputs, relative weights describe the intended selection proportions, with the integer modulo sampling used by the code. The block-derived input limitations in Section 9 still apply; weighting does not make the process cryptographically unbiased.

Anchored Random Walk NFT selections remain separate. They select among active NFT anchoring records, so an address with more eligible NFTs has more entries. They do not use the participant ETH weights and carry no ETH allocation.

6.5 Retrieval and expiry

Secondary ETH and attached assets have retrieval rules implemented by the Allocations Wallet. At cycle end, the wallet stores a deadline using its configured timeout, currently five weeks. The designated recipient or cycle beneficiary has priority before expiry. At or after expiry, another caller can retrieve eligible uncollected assets for themselves. Expiry grants permission; it does not automatically send the assets.

ETH balances are tracked separately for each cycle and recipient. Multiple allocations to an address in one cycle accumulate in that cycle's balance; an allocation in a later cycle does not extend an earlier cycle's deadline. Attached assets also use their cycle's stored deadline. Changing the timeout setting affects later cycle-end registrations, rather than rewriting deadlines already recorded. Always identify the wallet that holds a historical allocation; replacing the wallet does not migrate its balances.

7 Anchoring and coordination

7.1 Anchoring NFTs

Anchoring transfers an NFT into the relevant wallet. Only the recorded original staker can release that anchor. Each NFT can be anchored only once in that wallet over its lifetime. Releasing it returns the NFT but does not restore its eligibility, even after ownership changes.

Cosmic Signature NFTs share each successful ETH Anchor Distribution equally across the active NFT records. The accounting accumulates distributions per NFT; the participant receives the accrued ETH when releasing the anchor. Failed ETH delivery or NFT return reverts the release. Integer rounding can leave residual ETH. Once no NFTs remain anchored, the wallet owner can send its remaining ETH to a chosen nonzero address, including its own; owner powers therefore do extend to this residual balance.

Random Walk NFT Anchoring gives eligibility for the separate NFT/CST selections and does not accrue ETH distributions. Its eligibility record is independent of the protocol's once-per-NFT ETH cost reduction.

7.2 The Cosmic Council

CST uses timestamp-based voting checkpoints. Holding CST and delegating its voting power are separate actions; delegation can be to oneself or another address. Voting uses the applicable historical checkpoint rather than simply reading the current transferable balance.

The checked Council configuration uses a 100 CST proposal threshold, a two-day voting delay, a two-week voting period, and a quorum of 3% of total CST supply at the proposal snapshot. Support must exceed opposition, and support plus abstention count toward quorum. The Governor has no separate execution timelock; after voting ends, a successful proposal can execute directly. It can change its own governance settings through proposals. The Governor can execute successful proposals against assets and permissions that it actually controls. It does not acquire ownership of the protocol or its wallets merely because it exists.

7.3 Public Goods and Outreach

At the snapshot, the protocol's Public Goods destination is a vault whose configured beneficiary is the Protocol Guild address listed in Appendix A. Finalization sends ETH to the vault; a separate permissionless vault operation forwards its accumulated balance to the beneficiary. This is not an automatic same-transaction transfer from every cycle to the final beneficiary.

The vault owner can change the beneficiary. The Outreach Reserve receives its configured CST allocation, and its appointed treasurer distributes CST. The wallet owner appoints that treasurer. At the snapshot, the relevant owners and treasurer are the deploying address, not the Council. These are distinct administrative powers, not powers conferred on every CST holder.

8 The artwork

8.1 On-chain provenance and off-chain rendering

Each Cosmic Signature NFT stores a 32-byte seed at creation. Its `NftMinted` event records the cycle; the cycle is not a field in the token's stored data. The NFT contract uses Keccak-256 to derive each seed from a supplied counter, which starts from a value derived during finalization and increments for each NFT. It stores the resulting seed. This Keccak-256 step is distinct from the renderer's SHA3-256 stream below. The protocol does not enforce seed uniqueness; a hash-based construction does not constitute a uniqueness check.

The renderer initializes a SHA3-256 pseudorandom stream from the seed to explore gravitational three-body trajectories and select an orbit. Participation affects when finalization occurs and the chain context from which seeds are derived. Messages and individual Gesture amounts are not instructions for drawing particular features.

The description here uses generator revision `59b324d035252c7db503743441d04d0355b81f3b`. It is a separate software release from contract V3. Upgrading the contract does not install that renderer or prove that every hosted token has been regenerated with it.

8.2 Simulation and the main spectral edition

The generator's default search considers 100,000 candidates per batch with a configurable one-million-step setting for each of the warmup and recording phases. Candidates may be rejected before completing those phases. A fourth-order Yoshida integrator advances the three-body system. Selection uses Borda-style ranking across chaos, equilateralness, and spatial interest, followed by an image-space aesthetic stage over a shortlist of up to 24 candidates, with up to three retry batches.

The selected orbit is rendered through seed-dependent layers and camera parameters. Color is sampled from a continuous perceptual palette, including hue dispersion and variation; it is not fixed at 120-degree spacing. The main spectral edition uses 64 wavelength bins from 380 to 700 nanometers, with AgX tonemapping and configured finishing effects such as halation, prism, spikes, and stardust.

The package includes a 16-bit Display P3 master image, web derivatives, reveal films in web H.264 and archival HEVC formats, spectral images, a spectral sweep film, and metadata. A roughly 30-second main film is a target, with actual duration determined by frame count and schedule.

8.3 The Ember edition

The same selected orbit also drives the Ember edition: tidally shaped bodies move through a two-dimensional fluid, carrying black sumi ink over a modeled kozo surface. Ember follows the main edition's recorded view. Its optical model uses 36 spectral bands, distinct from the main edition's 64 bins.

The reviewed algorithm is `ember-v4`. Its outputs include a still, a normal film, a four-times-slower medium film, a ten-times-slower film, and a determinism certificate. The additional frames in the slower films are simulated, not interpolated. Ember can be omitted from a generation run, and a failure in that edition can leave the main package available without it; the asset manifest records the available outputs. Ember is an off-chain rendering edition of the same NFT; it does not create additional NFTs or alter a cycle's allocation count.

8.4 Reproducing a Signature

Reproduction requires the generator revision, configuration, orbit/view inputs where applicable, and relevant runtime information as well as the NFT seed. The main spectral renderer uses architecture-specific numerical paths that can differ slightly across machines. A promise of universal pixel identity for the entire pipeline would be too strong.

Ember uses controlled arithmetic and records SHA-256 digests of raw pixel streams, along with the inputs on which they depend. Its golden tests run across CPU architectures and thread configurations. That guarantee applies to the recorded Ember inputs and raw pixels, not universally to encoded PNG, MP4, or WebP bytes, which depend on encoder versions.

The contract and reviewed generator repositories carry CC0 licenses for their own code. Dependencies retain their own licenses. NFT metadata and generator URIs remain owner-configurable; on-chain seed provenance and the availability or immutability of hosted files are different properties.

9 Security assumptions and administrative powers

9.1 Randomness and transaction ordering

Finalization mixes the previous block hash, base fee, and available Arbitrum precompile values, including the previous Arbitrum block hash, gas backlog, and L1 pricing counters. Successive hashes of an incremented seed provide values for selections and NFT creation. Failed precompile reads omit that contribution and emit a diagnostic event.

This construction uses block-derived inputs. It is not a verifiable-randomness service or a commit-and-reveal protocol. Inputs can be predictable or influenced by transaction timing and block production. Calls within the same transaction can obtain the same initial seed; same-block values may be correlated. Hashing does not remove those trust assumptions. Timing, cost, and percentage formulas are deterministic, while selections and NFT seeds depend on this construction.

9.2 Contract and operational risks

Gesture and finalization paths use reentrancy protection, but not every external function is guarded. Escrow prevents secondary ETH recipients from blocking finalization through their receive functions; the direct main ETH transfer and other external dependencies still have failure paths. Token creation without receiver callbacks avoids one class of callback, while also allowing NFTs to be delivered to contracts that cannot later move them.

Owner-configurable values are not all range-checked. Invalid divisors, counts, percentages, or destination contracts can cause reverts or make operations unusable. The absence of a mid-cycle setter does not establish that all possible owner-selected configurations are safe. Participants also depend on Arbitrum execution and transaction ordering, timely finalization and retrieval, and off-chain availability for the rendered art.

The retained Allocations Wallet uses checked arithmetic when adding its configured timeout to the cycle-end timestamp. An excessively large timeout can therefore revert that registration and finalization. Replacing the Game implementation alone does not change this separately deployed wallet's code.

The Hacken report, published January 9, 2026, identifies reviewed commit `9385ddb`. That is historical evidence about its stated scope, not an audit of this V3 commit. Source verification establishes a relationship between source and deployed bytecode, not freedom from defects. The repository's older Certora material is explicitly historical and does not establish complete formal verification of V3.

9.3 Ownership and future governance

Most core configuration setters and the UUPS upgrade authorization require the owner and an inactive cycle, determined by the activation timestamp. Some controls have different rules: the next-cycle delay is adjustable at any time; activation can be changed before the first Gesture; ownership transfer and renunciation have their own inherited permissions. Companion-contract owners can separately change settings such as metadata references, the Public Goods beneficiary, the Outreach treasurer, and future retrieval timeouts.

The upgrade power is substantial. An owner can install different implementation logic when the current authorization conditions permit it. Constraints in today's implementation are not a guarantee that an authorized future implementation preserves them.

The previous paper described an intention to remove the deploying address's privileges after development. V3 contains no automatic sunset, scheduled renunciation, or compulsory Council transfer. Transferring ownership to the Council preserves the relevant powers under collective control; renouncing an ownership role disables that role's owner-only operations. They are different outcomes and must be handled for each contract separately. Any future change should be described as completed only after its transaction is observable.

10 Deployment history and the V3 transition

10.1 V1 launched the current proxy

The current proxy was deployed on **April 24, 2026 at 06:42:34 UTC**, in block 455767589. Its initial implementation was `0x7739148013777c485AD9f3d971e1005Eca686661`. The deployment transaction records that implementation. This is the launch date of the current proxy; references to a March launch in older repository text do not describe this transaction.

V1 introduced the cycles, ETH and CST participation, recognition tracks, selections, Anchoring, and supporting contracts. Its default Participation CST was a flat 100 CST paid to the participant making each Gesture. Its CST window duration was derived from the cycle increment. Overtime Gestures set the new deadline from the later of the old deadline and the current timestamp, then added an increment. The launch default for the main finalization exclusivity timeout was one day.

10.2 V2 changed timing and Participation CST

The V2 upgrade executed on **June 21, 2026 at 19:38:53 UTC**, in block 475913365. June 22 is the V2 implementation's source-verification date, not the upgrade date. V2 remains active at the October 17 snapshot.

Its principal changes were:

- **Elapsed-time Participation CST.** The flat amount became `floor(sqrt(floor(elapsed * multiplier / I)))`, paid to the current participant, in CST smallest units. V2 initializes the multiplier to `1.08 * 10^46`. At the original one-hour increment, this is approximately `sqrt(3 * elapsed seconds)` CST: about 13.416 CST after one minute and 103.923 CST after one hour. For a cycle's first Gesture, elapsed time starts at activation; a non-positive elapsed interval produces zero.
- **Minimum-amount protection.** Gesture methods gained a minimum accepted Participation CST argument.
- **Adaptive CST duration.** A stored duration begins at 12 hours on V2 initialization and changes with participation. At divisor 250, ETH Gestures reduce it by approximately 0.398%, while CST Gestures increase it by approximately 0.4%, subject to integer rounding. The exact updates are

`floor((duration + 1) * 250 / 251)` and `duration + floor(duration / 250)`, respectively. This duration carries between cycles instead of growing with the cycle increment as it did in V1.

- **Stored-deadline extension.** Later Gestures add to the existing deadline, even in overtime, rather than always putting it in the future.
- **Longer exclusivity.** The finalization timeout becomes 48 hours.
- **Next-cycle arithmetic.** Arithmetic in next-cycle preparation, including the scheduling addition, uses wrapping arithmetic instead of reverting on overflow. This addresses that particular overflow path; it does not make every parameter configuration safe.

V2 retained equal-weight participant entries, one main NFT, and the 25/8/4/6/7 ETH split for main, Chrono-Warrior, participant selections, Anchoring, and Public Goods respectively.

The verified V1 and V2 sources also show integration and storage changes:

- All six explicit Gesture methods, including the attached-asset variants, gained the minimum Participation CST argument and therefore changed function selectors. Direct plain-ETH transfers still call the ETH path with default arguments, including a zero minimum in V2. V2 adds reward quote getters and the `BidCstRewardAmountMinLimitNotReached` error.
- `BidPlaced` gained the Participation CST amount and updated CST duration, changing its event signature. Configuration getters, setters, and events changed from the fixed CST amount and duration divisor to the reward multiplier, stored duration, and duration-change divisor.
- V2 retained the proxy and existing populated storage positions. It reused the old CST duration-divisor slot for the stored duration and the fixed CST amount slot for its multiplier, and took one reserved slot for the new duration-change divisor. Its reserved gap was normalized without moving populated fields. `initializeV2()` replaced the V1 initialization entry point and initialized those three values plus the 48-hour timeout.
- The V2 implementation assumes an already-initialized proxy beyond its original cycle. It omits the original-cycle fixed-price branch and replaces the first-cycle guard with assertions used in diagnostic builds. It also checks activation before rejecting a CST opening Gesture, whereas V1 checked those conditions in the reverse order. Owner-only upgrade and configuration restrictions otherwise remain in place.

These implementation changes did not replace the companion contracts or change the Endurance Champion and Chrono-Warrior selection rules.

10.3 What V3 changes from deployed V2

Mechanism	Deployed V2	Proposed V3
Participation CST formula	Square root of scaled elapsed time	Linear scaled elapsed time
Participation CST recipient	Current participant	Previous participant
Opening Participation CST	Calculated from activation time	None
Late-cycle cost	No premium	Capped eighth-power premium under default settings
CST window adaptation	Stored duration, divisor 250	Decline rate, divisor 100, with duration safeguards
Configured CST beginning minimum	200 CST	1 CST plus the rate-based restart floor
Participant selection weights	One equal entry per Gesture	Contemporary ETH base cost
Main NFTs	1	3
Typical total NFTs	24	26
Typical fixed CST	27,000	27,000
ETH split, in the order above	25/8/4/6/7	20/15/5/5/5
Per-cycle data	Legacy participant/address records	Expanded `roundStats` and individual Gesture records

The reward and cost changes have distinct effects. Moving Participation CST to the previous participant recognizes the interval their Gesture remained most recent. The premium raises late costs without buying extra weight or inflating later base costs. Weighting recognizes the prevailing ETH base cost rather than counting every Gesture equally. The CST safeguards limit very short restarted windows while preserving zero-cost participation after enough elapsed time. These are mechanical effects, not guarantees about participant strategies, market prices, fairness, or eventual cycle duration.

V3 also changes integration details. `getDurationUntilMainPrize` returns a signed value, including negative overtime. Legacy duration setters remain callable by signature but revert; their stored getters are historical remnants. `BidPlaced` keeps its event signature while its eighth field changes from a duration to a decline rate, and its CST amount now belongs to the previous participant. `MainPrizeClaimed` gains an NFT count and changes signature. Several legacy getters are removed, most Game custom errors lose their text argument, and failure events become more specific. Consumers must distinguish historical V2 data from V3 data even when an event signature is unchanged. Use the ordered `Upgraded` event on the proxy as the version boundary. A V2 finalization can occur earlier in the same block as the V3 upgrade, so the block number alone is not sufficient to select the correct interpretation.

The V2-to-V3 technical change register covers these interfaces, storage, shared-source changes, rounding details, and failure semantics. A change to a retained companion contract's source does not update that deployed contract by itself.

10.4 Initializing V3 and handling historical assets

The owner upgrades the existing proxy with `upgradeToAndCall`, passing `reinitialize()` in the same transaction. The old implementation's authorization requires the owner and an inactive cycle. The reinitializer is versioned but is not independently owner-restricted, so initialization must not be left as a later public call.

Initialization sets the new pricing variables, linear CST multiplier, ETH percentages, and main NFT count. It overwrites the specified V2 configuration values. It preserves the cycle increment, the carried first CST beginning cost, existing timeouts, next-cycle delay, and component addresses. It emits the

initialization event rather than a separate change event for each rewritten setting. Clients must read the resulting configuration.

The expanded statistics reuse storage compatibly, but historical cycles do not gain data that was never stored. Their new fields are not retroactively populated, and legacy per-Gesture records have no V3 weights. The V3 `BidInfo` record contains an address and cumulative weight, not a timestamp. Complete Gesture timing must be reconstructed from events and their blocks; `biddersInfo` stores only each address's latest Gesture timestamp in each cycle. Historical event reconstruction must remain version-aware.

A new Allocations Wallet is an **explicit additional operator action**. The default upgrade does not replace it. If a new wallet is selected, the operator deploys it for the same proxy and sets the proxy's wallet address while the cycle is inactive. The old wallet retains its assets, deadlines, beneficiary records, and retrieval functions. The new wallet starts with empty records and a fresh attached-NFT index, so an index is meaningful together with its wallet address. Ownership and any customized timeout require separate configuration.

11 White paper revision history

11.1 The original English edition

The previous English white paper is version **1.0**, with **August 2026** on its title block and `2026-08-24` as its ISO date in the frontend source. The retained source snapshot is frontend commit `893368318d58a6c13add49336fd8fc2fb17ba45a`. That edition describes live V2 and an earlier V3 proposal. Its existing files and translations remain in their original repository as historical editions.

11.2 What changed in edition 1.3

Edition **1.3** is the canonical English paper in the Solidity repository. The suffix aligns with the proposed contract V3; paper and contract version numbers are separate. This revision follows the retained 1.0 edition and does not assert intervening published editions numbered 1.1 or 1.2.

Earlier paper section	Revision in 1.3
Abstract and design	Separates on-chain provenance from off-chain rendering; removes guaranteed reserve growth and automatic decentralization claims
Cycles and Gestures	Explains exact stored-deadline behavior, opening base cost, live delay versus default, and V3 pricing
Reserve and recognition	Updates ETH percentages, main NFT count, weighted selections, custody, conditional allocations, and retrieval deadlines
CST	Replaces the square-root/current-participant model with the linear/previous-participant model; explains opening and finalization exceptions
Art	Updates selection and palette descriptions, adds Ember v4, and separates the two editions' reproducibility guarantees
Anchoring and Council	Clarifies residual ETH, actual ownership, voting delegation, and Council authority
Public Goods	Distinguishes transfer to the vault from forwarding to its configured beneficiary
Security	Scopes historical reviews, corrects universal guard/failure claims, and states randomness and configuration assumptions
V3 proposal	Replaces the obsolete one-change, 10-times-premium proposal with the full reviewed V3 behavior
History and appendices	Adds chain-backed V1/V2 dates, V3 transition requirements, version comparisons, and a fixed-block deployment record

The earlier V3 expression `1 + 9 * (t / T)^8` described a 10-times total cost. The reviewed implementation instead produces approximately a five-times total cost at default settings, with the exact integer formula in Section 4.3. Likewise, the new main NFT count does not justify multiplying the main CST allocation or increasing typical fixed CST from 27,000 to 29,000.

Future editions should record the contract commit they describe, whether it has been deployed, and both protocol changes and corrections to the paper. A future translation should identify the English edition from which it was translated.

Appendix A Deployed contracts and source evidence

The following addresses and configuration observations refer to **Arbitrum One, chain 42161, block 515784460**, timestamp **October 17, 2026, 21:25:00 UTC**. The proxy points to V2 at that block. A future V3 implementation address is not assigned by this paper.

Component	Address
Protocol proxy	`0x6a714Ae7B5b6eA520F6BCA23d2E609C4Fd5863F2`
V2 implementation	`0x50eB3d05d2C463949DE9238D419385594f7AdB97`
CST	`0xAD91843e6A58Ba560F577E676986AFb1dba6FBA0`
Cosmic Signature NFT	`0xbb84Be3500A63581d3F2d5AC3bdF8685AAedad25`
Random Walk NFT	`0x895a6F444BE4ba9d124F61DF736605792B35D66b`
Allocations Wallet	`0xE1b619e9B39ea4109D2F429Ea5eAA307759b0011`
Cosmic Signature Anchoring	`0x6308A405B4FF1eA890870Efe2a6D036750B81F7C`
Random Walk Anchoring	`0x5EB3396092841E6c5b0b51141699F6711E830529`
Public Goods Vault	`0x96bB0ADB414d5350f435E52f94946B6C7A0760a9`
Outreach Reserve	`0xa3802c799f5e3D3D3562A9B513a41C6aAF92e25e`
Cosmic Council	`0xF3D52E1c681949be7E624778dB13DaD7F8c729db`

The Public Goods Vault's beneficiary is `0xdddd576bAF106bAAe54bDE40BCac602bB4a7cf79`. The protocol, NFT, Allocations Wallet, Public Goods Vault, and Outreach Reserve owner reads identify `0x14c82Ce4E5713E88C9462680E9c02BF4a3089871`; that address is also the checked Outreach treasurer.

The accompanying `deployment-snapshot.json` records the block hash, raw getter values, deployment transactions, runtime bytecode hashes, and Sourcify records. All listed runtime bytecodes matched those records at the snapshot. The 48 project sources in the verified V2 build match repository commit `3f52eaaf2258ec30f61fad07bb07cb2e3b4ca8a6` exactly. Companion contracts have their own deployment source records; current repository copies must not be assumed to reproduce every historical deployment.

Appendix B Important parameters and units

Parameter	V3 initialization or retained setting
ETH base-cost step divisor	100, retained
ETH opening floor divisor	200, retained
CST decline-rate change divisor	100, initialized
CST initial decline rate	16,666,666,666,666,667 smallest units per second
CST beginning-cost minimum	1 CST, initialized
CST restart duration safeguard	180 seconds, constant
Linear Participation CST multiplier	`6 * 10^25`, initialized
Late-window duration divisor	3,000,000, initialized
Late-premium base multiplier	29,228,998,656, initialized
Late-premium exponent	8, initialized
Cycle increment	3,709,083,600 microseconds at snapshot, retained
Cycle increment growth divisor	100, retained
Finalization exclusivity	172,800 seconds at snapshot, retained
Next-cycle delay	604,800 seconds at snapshot, retained
Allocations Wallet timeout	3,024,000 seconds at snapshot; separate wallet setting
Main NFT count	3, initialized
Fixed CST per recognition allocation	1,000 CST, retained
Outreach CST per cycle	3,000 CST, retained
Participant ETH / NFT selections	3 / 10, retained
Anchored Random Walk selections	10 when eligible records exist, retained
Gesture message limit	280 bytes, retained

Values retained from V2 can differ from initial launch constants. Mutable settings should be read from the intended deployment before preparing a transaction. The obsolete stored CST duration and its change divisor remain in storage for compatibility but no longer govern V3 pricing.

References

1. Reviewed V3 contract source, including pricing, finalization, initialization, and constants.
2. Verified V2 deployment record and sources.
3. Current proxy deployment transaction and V2 upgrade transaction.
4. Technical V2-to-V3 change register and endurance and Chrono-Warrior specification.
5. Reviewed art generator revision, including its Ember certificate implementation.
6. Hacken historical security report.
7. Previous English paper source.

Citation: Bobrovytsky, T. (2026). *Cosmic Signature: A procedural art protocol on Arbitrum One.* White paper 1.3, canonical English edition.

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